SALES & PROFITABILITY ANALYSIS

Sample Superstore Sales Dataset
Tableau Dashboard Project Report

14.92M	1.52M	5,496	8,399
Total sales	Total profit	Distinct orders	Order-line records

Prepared from the uploaded Excel workbook (Orders, Returns and Users sheets), the Tableau packaged workbook, and the supplied five-page dashboard PDF.

Analysis period: 01 Jan 2009 to 30 Dec 2012

1. Executive Summary

The supplied Orders sheet contains **8,399 order-line records** representing **5,496 distinct orders** from 2009 to 2012. Across the period, recorded sales total **14,915,601** and recorded profit totals **1,521,768**, giving an overall profit-to-sales ratio of **10.20%**. The dataset records 214,777 units of order quantity.

Regional picture: Central is the largest sales region at 4,699,167 (31.51% of sales) and also records the highest regional profit (481,891). South has the strongest regional profit margin (13.41%), while West has the lowest (8.21%).

Product picture: Technology contributes the most sales (5,984,248) and the most profit (886,314). Furniture has a much thinner profit margin (2.27%) than Technology (14.81%) or Office Supplies (13.80%).

Customer picture: Corporate is the largest customer segment by sales (5,498,905) and profit (599,746). **Monthly quantity:** May is the peak month (21,273 units) and November is the lowest (16,251 units).

Important interpretation: the dashboard is descriptive, not causal. The findings identify where sales and profit are concentrated and where losses occur; they do not establish why those patterns occurred.

Headline KPIs

Metric	Result	Definition
Sales	14,915,601	Sum of Sales across all order-line records
Profit	1,521,768	Sum of Profit across all order-line records
Profit-to-sales ratio	10.20%	Total Profit divided by Total Sales
Order quantity	214,777	Sum of Order Quantity
Distinct orders	5,496	Unique Order ID count
Distinct customers	795	Unique Customer Name count
Average order sales	2,713.90	Mean of order-level sales totals
Median order sales	899.43	Median of order-level sales totals

2. Dataset Description and Preparation

The project uses the classic Sample Superstore Sales workbook supplied with the request. The Tableau workbook identifies the Orders table as the primary data source and lists 23 fields, including order dates, sales, profit, shipping, customer, region and product attributes.

Workbook sheets

Sheet	Observed size	Role in this analysis
Orders	8,399 rows × 25 columns	Main transactional dataset used for all sales, profit, quantity, region, segment, category and time analys
Returns	572 rows × 2 columns	Return-status lookup; contains 572 distinct Order IDs marked Returned.
Users	8 rows × 2 columns	Region-to-manager lookup; 8 rows in the supplied workbook.

Important Orders fields

Field group	Fields	Analytical use
Identifiers & dates	Row ID, Order ID, Order Date, Ship Date	Record identity, distinct order counts, annual/monthly analysis and shipp
Measures	Order Quantity, Sales, Profit, Discount, Unit Price, Shipping Cost, Product Base Margin	Core sales metrics and profitability analysis.
Customer & geography	Customer Name, City, Zip Code, State, Region, Customer Segment	Customer, location and regional comparisons.
Product & fulfillment	Product Category, Product Sub-Category, Product Name, Product Container, Ship Mode, Order Priority	Product, fulfillment and order-priority analysis.

Preparation and quality checks

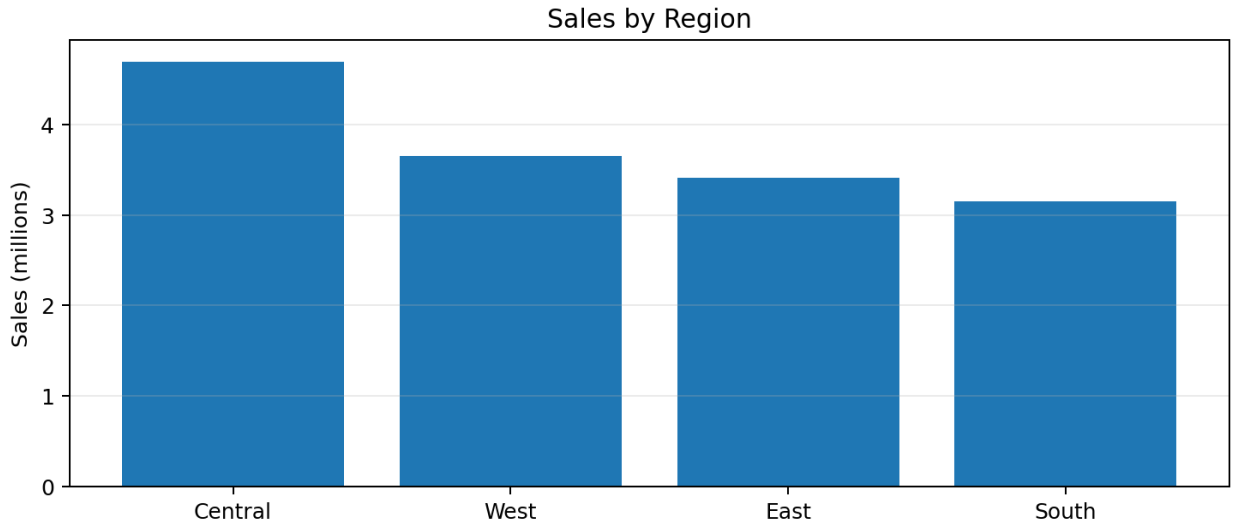
• Parsed Order Date and Ship Date as dates and derived calendar year and month from Order Date.
• Used Order ID distinct count for order totals; the 8,399 Orders rows are order lines, not 8,399 separate orders.
• Checked missing values: 63 records have a missing Product Base Margin; the other 22 Orders fields have no missing values in the supplied sheet.
• Matched Returns Order IDs against Orders: all 572 return IDs occur in Orders, equal to approximately 10.41% of distinct orders. This is an order-level return share, not a returned-item or returned-revenue rate.
• Sales and Profit are treated as numeric values in the dataset's own units. The workbook does not explicitly specify a currency code, so this report does not label the values as USD or INR.

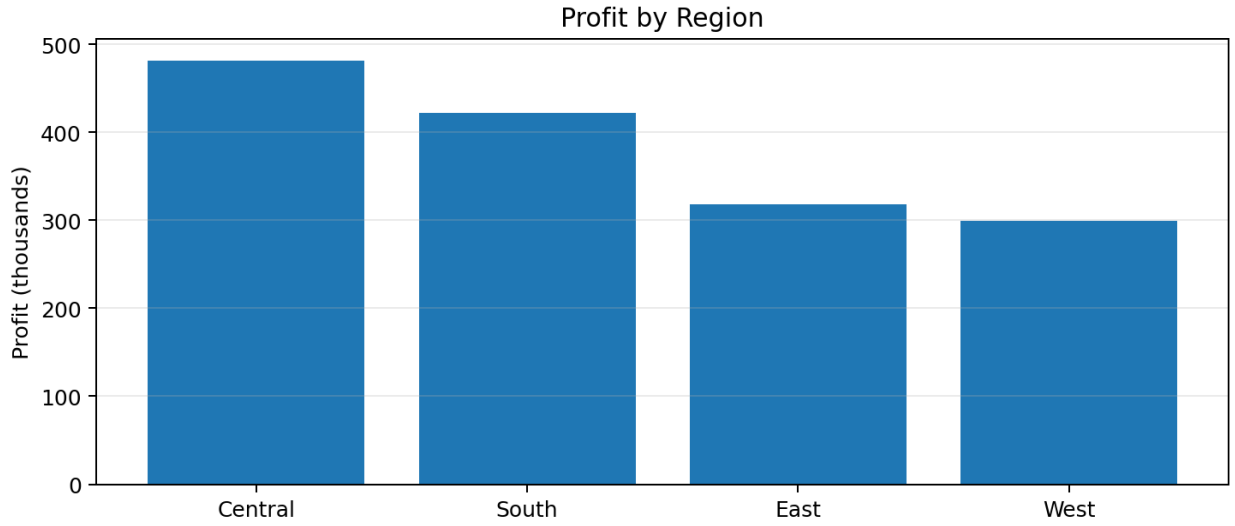
Calculation notes

Total Sales = sum(Sales); Total Profit = sum(Profit); Profit-to-sales ratio = Total Profit ÷ Total Sales × 100; regional sales share = region Sales ÷ Total Sales × 100. Monthly order quantity is the sum of Order Quantity grouped by the calendar month of Order Date, combining all years in the dataset.

3. Regional Performance

The dashboard's regional sales contribution chart and the workbook calculations agree to rounding. Central is the strongest region on both sales and total profit. South ranks fourth in sales but second in profit, which indicates a stronger profit return relative to its sales than the other regions.



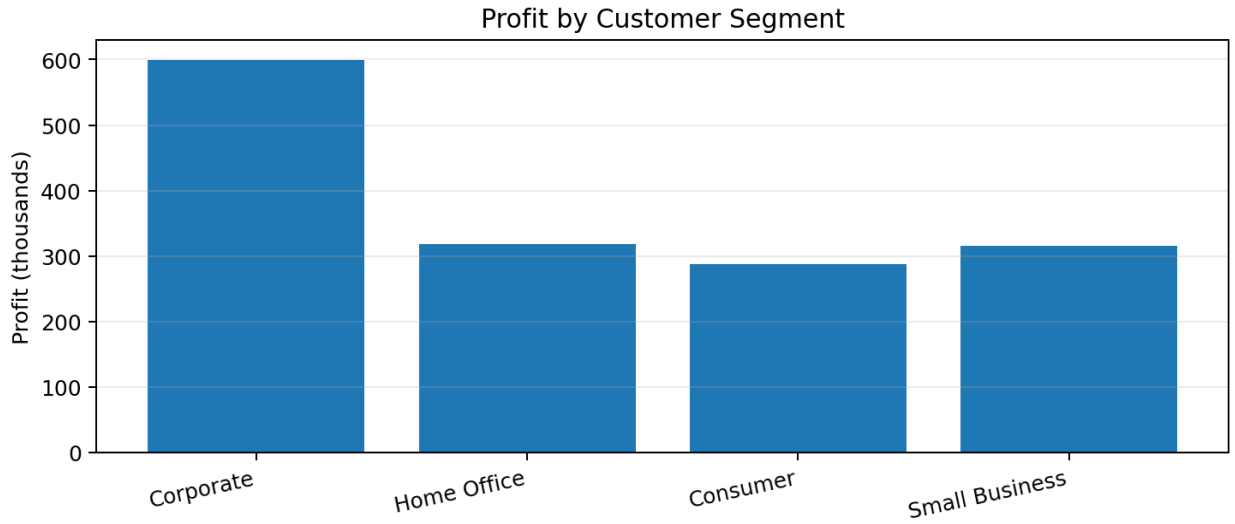


Region	Sales	Sales share	Profit	Profit margin
Central	4,699,167	31.51%	481,891	10.25%
West	3,649,748	24.47%	299,518	8.21%
East	3,416,466	22.91%	317,852	9.30%
South	3,150,219	21.12%	422,507	13.41%

Interpretation: Central generates the largest absolute sales and profit. South's margin is highest at 13.41%, suggesting its sales mix is more profitable in aggregate. West contributes 24.47% of sales but has the lowest regional margin (8.21%), so a review of product mix, discounting, shipping costs and loss-making order lines in West is warranted.

4. Customer Segment Analysis

The Tableau dashboard displays profit by Region and Customer Segment. The overall workbook totals show Corporate as the largest segment by both sales and profit. The segment chart below complements the regional stacked bars by aggregating each segment across all regions.



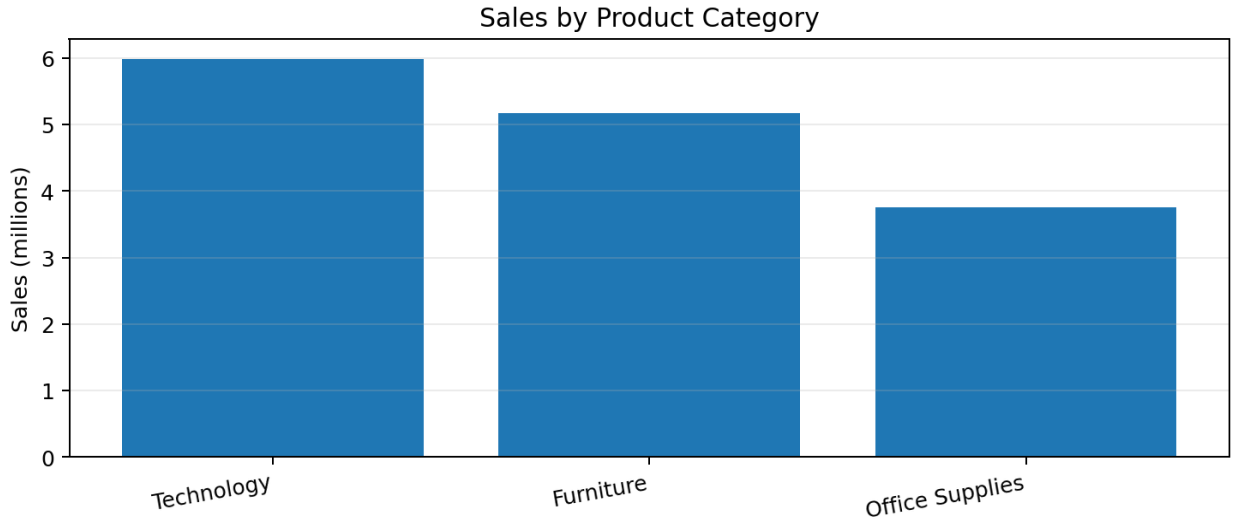
Customer segment	Sales	Profit	Order quantity
Corporate	5,498,905	599,746	78,517
Home Office	3,564,764	318,354	52,807
Consumer	3,063,611	287,960	41,760
Small Business	2,788,321	315,708	41,693

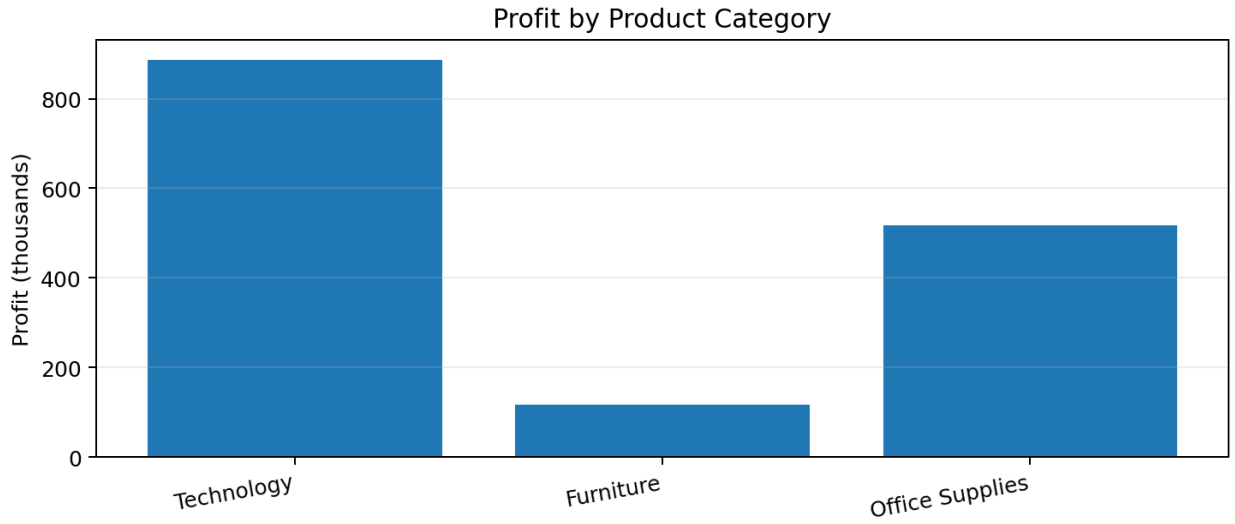
Regional segment observations from the dashboard: Corporate is the largest profit contributor in Central (196,081), East (116,516), and South (168,025). In West, Corporate profit is 119,123 and Small Business profit is 92,794. The Consumer segment is particularly material in South (131,406), where it is the second-largest segment contribution.

Implication: retain and understand the Corporate segment's contribution while also examining regional differences. Segment totals alone do not show customer acquisition cost, retention, or lifetime value; those measures are not present in this workbook.

5. Product Category and Sub-Category Analysis

Technology leads the dataset in sales and profit. Office Supplies has the largest total order quantity, while Furniture produces substantially lower profit relative to its sales.





Category	Sales	Profit	Profit margin	Quantity
Technology	5,984,248	886,314	14.81%	52,175
Furniture	5,178,591	117,433	2.27%	44,324
Office Supplies	3,752,762	518,021	13.80%	118,278

Sub-category risk and opportunity

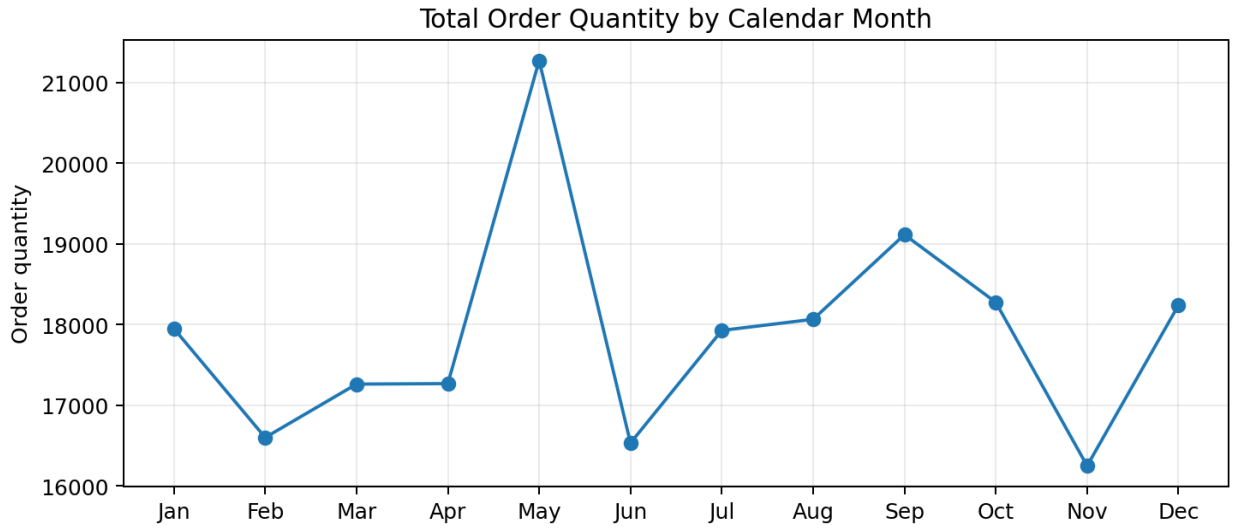
Lowest-profit sub-categories	Profit
Tables	-99,063
Bookcases	-33,582
Scissors, Rulers and Trimmers	-7,799
Rubber Bands	-103
Highest-profit sub-categories	Profit
Telephones and Communication	316,952

Lowest-profit sub-categories	Profit
Office Machines	307,713
Binders and Binder Accessories	307,413

Key finding: Tables has the largest sub-category loss (-99,063) and Bookcases is also loss-making. These should be reviewed for pricing, discount levels, product cost and shipping expense before expanding sales. The data does not prove that any one factor caused the losses.

6. Time Trends and Sales–Profit Relationship

The dashboard presents monthly order quantity from January to December. The workbook values confirm that May has the highest combined quantity and November the lowest. Because the chart combines all four years, it describes seasonality by calendar month rather than a year-by-year trend.



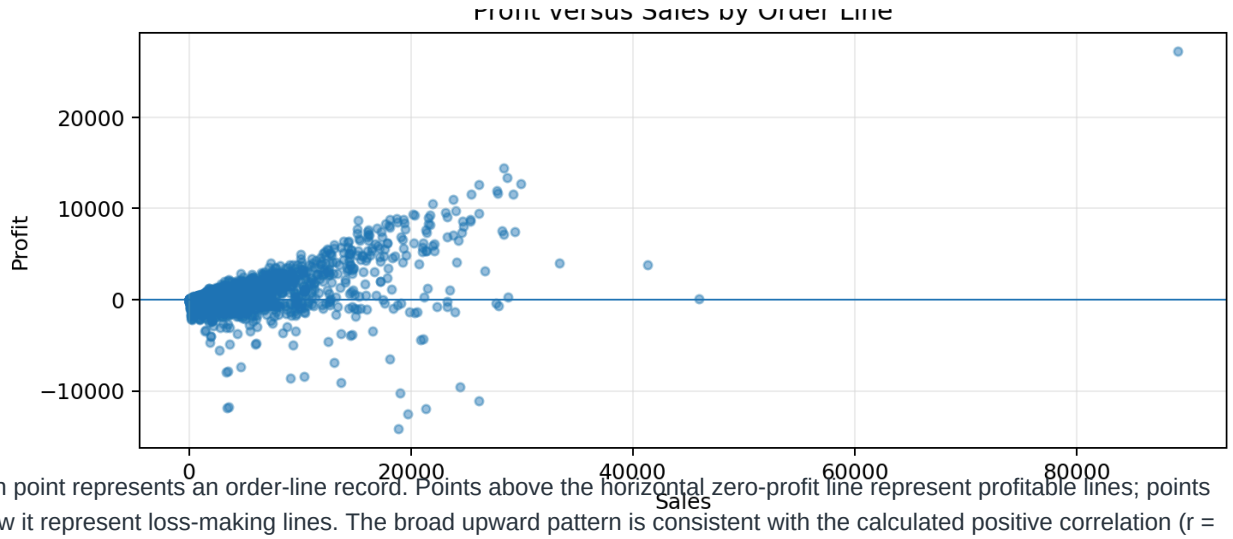
Month	Order quantity
January	17,952
February	16,602
March	17,264
April	17,270
May	21,273
June	16,534
July	17,929
August	18,067
September	19,116
October	18,278
November	16,251
December	18,241

Annual sales and profit

Year	Distinct orders	Sales	Profit
2009	1,394	4,209,139	434,539
2010	1,393	3,549,681	363,871
2011	1,317	3,436,817	381,456
2012	1,392	3,719,964	341,902

The Profit vs Sales scatter plot in the supplied dashboard shows a generally positive association: higher-sales order lines often have higher profit, but there are also substantial negative-profit points and high-sales points with relatively modest profit. The order-line Sales–Profit correlation in the workbook is 0.58; correlation is descriptive and does not

Profit versus Sales: Reading the Scatter Plot



Each point represents an order-line record. Points above the horizontal zero-profit line represent profitable lines; points below it represent loss-making lines. The broad upward pattern is consistent with the calculated positive correlation (r = 0.58), but the spread of points shows that higher sales do not guarantee higher profit. A number of substantial sales lines have low or negative profit, supporting closer review of pricing, discounts, product cost and shipping expense.

7. Shipping, Returns and Business Recommendations

Shipping mode

Ship mode	Sales	Profit	Profit margin
Delivery Truck	6,224,879	269,645	4.33%
Express Air	1,184,419	147,430	12.45%
Regular Air	7,506,303	1,104,693	14.72%

Regular Air contributes the most sales and profit in absolute terms, and it has the highest profit margin among the three shipping modes. Delivery Truck has the lowest margin in this dataset. This comparison is descriptive; shipping mode selection may be influenced by order size, product type and destination.

Returns

The Returns sheet contains 572 unique Order IDs, each marked Returned. All are present in the Orders sheet. This is 572 returned orders out of 5,496 distinct orders (10.41%). The workbook does not provide a return reason or returned quantity/value, so financial impact and causes of returns cannot be calculated from these files.

Recommendations

1. Investigate West region profitability: it has the lowest regional margin (8.21%) despite being the second-largest region by sales.
2. Review Furniture, especially Tables and Bookcases, for pricing, discounts, cost structure and shipping expense; avoid using sales volume alone as a success measure.
3. Protect the strong Technology category contribution and examine which products and customers drive its high profit.
4. Use May's higher order quantity to plan inventory and fulfillment capacity; examine November's lower quantity when planning seasonal demand.
5. Review Delivery Truck economics and the mix of orders assigned to each shipping mode before changing service policies.
6. Add return reasons, returned units and returned revenue in future data collection to understand return drivers and quantify their impact.
7. Monitor negative-profit order lines and orders routinely, since 4,264 of 8,399 order lines (50.77%) have negative line-level profit; at order level, 2,651 of 5,496 orders (48.24%) have negative total profit.

8. Conclusion, Limitations and Source Notes

Conclusion

The Sample Superstore analysis records 14,915,601 in sales and 1,521,768 in profit across 5,496 distinct orders during 2009–2012. Central is the largest region by sales and profit, South has the strongest regional margin, Corporate is the leading customer segment, and Technology is the leading product category by both sales and profit. The clearest profitability review areas are West region performance and the Furniture sub-categories Tables and Bookcases. Monthly order quantity peaks in May.

Limitations

• The data covers the period 1 January 2009 to 30 December 2012; it should not be interpreted as current market performance.
• Currency is not explicitly identified in the supplied workbook; values are shown in the dataset's recorded units.
• Product Base Margin is missing in 63 Orders records. Those rows were retained for Sales and Profit calculations because those fields are present.
• Returns data gives returned order IDs but not return reasons, units or monetary amounts.
• The monthly quantity chart aggregates all years together, so it cannot show whether a particular month rose or fell from one year to another.
• The supplied dashboard PDF is a static export. Interactive filters and underlying Tableau calculations were not independently tested; displayed totals were cross-checked against the workbook.

Source files used

1. Sample+-+Superstore+Sales+(Excel) (3).xls — Orders, Returns and Users worksheets.
2. Tableau Dashboard 1.twbx — packaged Tableau workbook; the workbook metadata identifies the Orders data source and the fields used.
3. Dashboard .pdf — five-page dashboard export showing Profit by Region & Customer Segment, Sales Contribution by Region, Profit vs Sales, and Order Quantity by Month.

Appendix: dataset field inventory

Orders sheet: Row ID, Order ID, Order Date, Order Priority, Order Quantity, Sales, Discount, Ship Mode, Profit, Unit Price, Shipping Cost, Customer Name, City, Zip Code, State, Region, Customer Segment, Product Category, Product Sub-Category, Product Name, Product Container, Product Base Margin, Ship Date.

End of report